

YIMIAO YU

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EDUCATION

New York University, Center for Data Science

New York, NY

Master of Science in Data Science; GPA: 3.78/4.0

Sep 2025 – May 2027 (Expected)

- Coursework: Machine Learning, Optimization & Computational Linear Algebra, Probability & Statistics, Big Data; upcoming (Fall 2026): Deep Learning, Natural Language Processing

Zhejiang Ocean University

Zhoushan, China

B.S. in Data Science & Big Data Technology; GPA: 3.69/5.0 ($\approx 86.9/100$)

Sep 2021 – Jun 2025

- Coursework: Algorithm Design & Analysis, Machine Learning, Data Visualization, Applied Multivariate Statistics

RESEARCH EXPERIENCE

Research Assistant — Random-Label Memorization & Transfer in Deep Networks

New York, NY

NYU Center for Data Science, Advisor: Prof. Florentin Guth

Jan 2026 – Present

- **Research question:** deep networks can memorize entirely random labels, yet how this memorization is organized in the parameters and whether it transfers to new tasks is poorly understood — this work characterizes it from a **transfer** perspective. Co-author of a working paper on transfer between memorization tasks (under review at a NeurIPS 2026 workshop); responsible for the transfer-effects results
- Discovered and characterized a **warm-start transfer acceleration**: pretraining on one random-label assignment (seed 21) significantly speeds up downstream memorization of a *different* assignment (seed 42); quantified onset via a delayed-sigmoid fit (midpoint t_0) and found downstream t_0 decreases with pretraining time, with a region where t_0 has slope ≈ -1 in pretraining epochs
- Showed the effect is **not tied to natural images**: pretraining on random pixels + random labels still accelerates downstream CIFAR-10 + random-label memorization
- Identified the **necessary condition** for transfer: resampling the label mapping every epoch (same image, different label) destroys the speedup, whereas resampling random-pixel inputs under a fixed label mapping preserves it — transfer requires a consistent input–output mapping, not repeated exposure to identical pairs
- Localized the effect via **ablations** (reset / freeze of BN affine parameters γ , β , FC layers, and convolutional groups): acceleration is carried mainly by **convolutional weights** (not BN affine or FC); no single conv group accounts for it, and block-3 pretrained weights are in fact harmful downstream
- *Ongoing*: SVD spectral analysis of convolutional weights, with controlled “replace Σ vs replace U/V ” experiments to test whether the spectrum shape or the singular vectors drives the transfer

PROJECTS

Gender Bias in Student Evaluations of Teaching | Python, Pandas, SciPy, Scikit-learn

Dec 2025

- **Question:** do student evaluations show systematic gender bias? Screened 18,422 instructors (≥ 5 ratings) from 89,893 RateMyProfessor records; tested with Mann-Whitney U, Levene’s, bootstrap CIs, and regression
- **Findings:** a small but statistically significant pro-male bias (Cohen’s $d = 0.064$, 95% CI [0.035, 0.092]) and higher female variance (0.914); tag-based rating model $R^2 = 0.71$ and a “pepper” logistic classifier AUROC = 0.79

Movie Ratings Hypothesis Testing | Python, Pandas, SciPy, Matplotlib

Nov 2025

- **Question:** do popularity, release year, gender, and viewing preferences affect movie ratings? Tested on 400 movies rated by 1,000+ users using Mann-Whitney U (Bonferroni-corrected), Kruskal-Wallis, and rank-biserial / ϵ^2 effect sizes
- **Findings:** higher-popularity films rated significantly higher ($|r| = 0.252$); inconsistent quality in 7 of 8 classic franchises

TECHNICAL SKILLS

Programming: Python (primary), SQL, Java, C

ML/Data: PyTorch, NumPy, Pandas, Scikit-learn, Matplotlib, SciPy

Tools: Git, Jupyter

Languages: English (IELTS 7.0, GRE 324), Mandarin (native)